

Mark Park

Lead Software Engineer

La Habra, CA | (310) 617-6737 | markhpark619@gmail.com | linkedin.com/in/markhpark | mark-park.com

SUMMARY

Lead Software Engineer with 4+ years on Franklin Templeton Wealth Management Technology's client onboarding and money-movement platforms. Founding engineer and backend lead of the Account Lifecycle Platform (AOP 2.0) rebuild for the High Net Worth business — a 5-engineer team delivering account opening, KYC/AML screening (LexisNexis), DocuSign e-signature workflows, Drools rules, and a dynamic workflow orchestration engine I designed to replace Camunda. Earlier a core builder and one of two primary production maintainers of CashApp, the firm's money-movement platform. Java/Spring microservices and Angular on Azure (Cosmos DB, SQL Server, Snowflake); established an AI-First, TDD-gated agent-based code generation practice on GitHub Copilot CLI adopted across three teams.

TECHNICAL SKILLS

Languages: Java 8/17/21 (led multi-version migrations), Python (Streamlit, internal tooling and automation), SQL, TypeScript, JavaScript (ES6), C++

Backend & Frameworks: Spring Boot, Spring MVC, Spring Security, Spring Cloud (Eureka, Feign, API Gateway), Spring Batch, Spring WebFlux, Hibernate/JPA, Lombok, REST APIs, API design (API-first), gRPC, OpenAPI/Swagger, Microservices, distributed systems, Domain-Driven Design, SAGA Choreography, Event-Driven Architecture

Client Onboarding & Account Lifecycle: Account opening and onboarding workflows, dynamic workflow orchestration (custom Spring/SQL Server engine; Camunda/BPMN), role-based approval routing across Front Office, Compliance, and Operations, Drools rule engine (BRMS) for real-time validation and automated decisioning, KYC/AML, sanctions, and risk screening API integration (LexisNexis; Instant ID evaluation), DocuSign e-signature and document collection workflows, Azure Blob document storage, data capture and validation pipelines, regulatory workflows and audit trails, OAuth2/OIDC and token/role-based authorization, digital-first client experience

Money Movement & Payments: ACH, wires, check; NACHA, wire, and positive pay files; Axway file integrations; FIS Global Plus accounting, Cash Agent, AML monitoring, and OFAC screening integrations

Front-End & Full Stack: Angular (production UI development alongside Spring services), React, TypeScript

Azure & Cloud: Cosmos DB, Service Bus, Event Grid, Functions, Key Vault, Blob Storage, API Management, Entra ID (Azure AD), Application Insights, Log Analytics (KQL); AWS (EC2, S3, ECS); Kubernetes-based deployments, Docker, RabbitMQ, Kafka, Redis; hybrid cloud/legacy environments

DevOps, Release & Observability: CI/CD (GitLab, Bamboo, GitHub Actions, Azure Pipelines, Jenkins), SonarQube, Datadog, integration API and workflow monitoring, ServiceNow incident/change tracking, production support, TQA release sign-off cycles, Maven, Gradle, Git

Databases: Azure Cosmos DB and MongoDB (NoSQL / non-SQL), SQL Server (MSSQL, stored procedures), Snowflake, MySQL

AI Fluency — AI-First Engineering & Business Workflow: Functional integration between applications and existing agents (GitHub MCP for agent-authored pull requests and code review; Copilot CLI commands integrated with Jira); generating high-quality code by using prompts efficiently (prompt and loop engineering, token-usage optimization); validating AI-generated outputs before applying to critical systems (TDD-gated, agent-reviewed changes); writing automation scripts and test cases; debugging complex transaction and onboarding flows; identifying opportunities to use AI; authored reusable copilot-instructions.md standards; led AI adoption across three product teams. Tools: GitHub Copilot (IntelliJ, Copilot CLI, multi-model), GitHub MCP (Model Context Protocol), Claude Code (self-taught on personal time)

Testing & Process: Automated testing of onboarding and transaction flows (JUnit, Mockito, JaCoCo), regression and API test automation, Postman, code peer review and branching strategy, technical documentation, root cause analysis, incident management against business-critical SLAs, full Agile SDLC, Jira, Scrum; working knowledge of PostgreSQL, Kotlin, Go, GraphQL

PROFESSIONAL EXPERIENCE

Brillio — Client: Franklin Templeton (Fiduciary Trust) | Remote (Eastern hours) — New York, NY

Contract Software Engineer (May 2022 – Jun 2025) → Software Development Lead, Brillio full-time (Jun 2025 – Present)

Embedded 4+ years on Franklin Templeton Wealth Management Technology teams; performance earned a two-year contract extension, conversion to full-time Brillio employee, and promotion to Software Development Lead, with recognition from directors, architects, and other teams' leads for the AOP 2.0, SWP Forms, and CashApp releases. California-based on Eastern hours; open to a three-day hybrid schedule at a Franklin Templeton location.

Lead Software Engineer — Account Lifecycle Platform (AOP 2.0, High Net Worth Client Onboarding) | Aug 2024 – Present

- Founding engineer on the rebuild; promoted to lead backend engineering on the recommendation of senior architects and management. Led up to 5 engineers taking a heavily criticized legacy onboarding platform from POC to MVP to production — a release leadership recognized as a turnaround.

- Automated over 60% of previously manual onboarding and KYC logic and improved responsiveness 60–80% over the legacy platform, reducing manual touchpoints and improving operational efficiency for onboarding, compliance, and operations teams.
- Designed and built the platform’s dynamic workflow orchestration engine (Spring Boot + SQL Server). Built the Camunda POC end-to-end with the IT manager; when a major Camunda upgrade broke the code months before MVP, rebuilt the entire workflow as a custom engine in one month. Its orchestration APIs cover every account-opening scenario — request, validation, assignment, Front Office / Retirement Services approval, KYC approval, DocuSign signature, SWP submission approval — with token- and role-based authorization against request status and sub-status, retries with backoff, and fault tolerance so no request ends in a broken state; new steps and approval paths are configured in the database without code changes.
- Architected the two core microservices along clear domain boundaries: a Drools-based rules engine governing all UI field behavior, with dynamic completion validation synced to the Angular UI for real-time, millisecond feedback, and a secure persistence service as the sole authorized gateway to internal Franklin Templeton systems.
- Delivered the document collection and e-signature workflow: UI and backend validation of required documents by account type and client, document storage APIs on Azure Blob, and production support of the DocuSign integration, including resolving live e-signature incidents.
- Built the KYC/AML screening API against LexisNexis from vendor documentation — a custom search returning screening hits for compliance review — as the POC for the onboarding KYC integration; contributing to the Instant ID evaluation and assisting the downstream IPS and KYC applications.
- Owned the data layer: Cosmos DB primary store (generic recursive Patch-API save utility, 5–10 ms, working around a known SDK upsert limitation), SQL Server workflow state, and Snowflake. Built a Python/Streamlit application on Snowflake that converts deprecated AOP 1.0 data into business-ready documents, used weekly by the business to review and migrate legacy accounts.
- Established an AI-First, TDD-gated agent-based code generation workflow on GitHub Copilot CLI: test suites written from business requirements first, AI agents implement against them, and every change is validated by agent-driven code review and agent-authored pull requests via GitHub MCP before it reaches production — existing functional tests are never broken. Self-taught the workflows on Claude Code on personal time, then translated them onto the firm-approved Copilot CLI.
- Led AI adoption across three product teams: authored the shared copilot-instructions.md, Copilot CLI commands integrated with Jira, and a Python database-alignment tool adopted by other projects; ran knowledge-transfer sessions on the CLI approach (prompt and loop engineering, reducing token usage, optimizing agent efficiency) for all 5 AOP 2.0 engineers plus the CashApp and SWP Forms leads. Top 5% firm-wide on Copilot cost-efficiency since the Jan 2026 rollout; 30K+ lines of AI-assisted code in production.
- Automated testing of onboarding flows with JUnit and Mockito; partnered with TQA on workflow-scenario automation scripts. Led the Java 17→21 migration ahead of MVP; sole backend on-call during market hours; grew into full-stack delivery of Angular features; mentored a summer intern to shipping production APIs.

Backend Engineer — SWP Forms (Systematic Withdrawal Plan) | Aug 2024 – Dec 2025 (concurrent with AOP 2.0)

- Hand-picked by the lead architect to build a standalone Drools rules-engine microservice from concept through MVP in 3 months; cut processing time for 500+ business rules from several seconds to 50–100 ms using fact injection, agenda groups, and salience-based prioritization. Later extended to a second production application; after handoff, onboarded the successor engineer and resolved production issues through Dec 2025.

Backend Developer — CashApp (Wealth Management money-movement platform: ACH, Wires, Check) | May 2022 – Aug 2024

- Core contributor from POC through MVP to production on the High Net Worth business’s money-movement platform: ACH, wire, and check processing, NACHA, wire, and positive-pay file generation and parsing, and Axway and Azure Blob file integrations.
- One of two primary production maintainers overseeing live-market-hours processing of tens of millions of dollars daily (\$106B+ lifetime) against strict settlement cut-offs — never rolled to the next business day; owned root cause analysis of production incidents against SLAs and TQA release sign-off. Owned OFAC screening, AML monitoring, compliance-reconciliation, and audit-trail processing; built FIS Global Plus accounting, Cash Agent, external payment, and Azure check-reading AI integrations.
- Hand-built the platform’s Spring/SQL Server transaction workflows and designed a scalable event-driven architecture on Azure Event Grid powering 20+ monitored scheduled batch jobs, several on 5-minute cycles. Eliminated production deadlocks (2–5 per day to zero) via optimistic concurrency control, retryable services, and custom thread pool executors; SAGA choreography over RabbitMQ, later Azure Service Bus.
- Delivered on Azure with Kubernetes-based deployments, GitLab/Bamboo CI/CD, regression and API test automation, and ServiceNow incident and change tracking; led design discussions across cross-functional groups of 20–30; later assisted the team’s Java 8→21 upgrade.

EDUCATION & CERTIFICATIONS

San Diego State University — Bachelor of Science, Computer Science

In progress: Anthropic Claude Certified Developer – Foundations; Microsoft Azure AI & Agents Developer Associate